

Mingjing (Marcus) Xu

Ph.D. Student | Swansea University

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Research interests: Multi-task and multi-objective learning; multimodal representation learning; language modeling; security and privacy in wireless sensing.

Conference reviewer: AAAI, AISTATS, ICLR, ICML, NeurIPS, and etc.

Education

Swansea University, UK

Ph.D. student, Faculty of Science and Engineering

Sep 2025 – Present

Rochester Institute of Technology, USA

Research Assistant | GPA: 3.94/4.00

Jan 2024 – Jan 2025

Temple University, USA

Teaching Assistant, CIS 3515: Introduction to Mobile Application Development | GPA: 3.89/4.00

Jan – Dec 2023

The Chinese University of Hong Kong, Hong Kong

M.Sc. in Electronic Engineering; GPA: 3.275/4.000; Supervisor: Prof. Hongliang Ren

Aug 2021 – Nov 2022

Shanghai University, China

B.Eng. in Electrical Engineering and Automation | GPA: 3.14/4.00 (83.30/100)

Sep 2015 – Jul 2019

Selected Publications

*Equal contribution | Venue rankings: 2026

- [1] Y. Zhou*, J. Shi*, M. Xu*, Z. Jiang, and J. Chen. Beyond Student: An Asymmetric Network for Neural Network Inheritance. *ICLR 2026*. [ICORE A*]
- [2] Y. Zhou*, M. Xu*, J. Shi*, Q. Li, and J. Chen. Collaborative Representation Learning for Alignment of Tactile, Language, and Vision Modalities. *AAAI 2026*, 40(22):18864–18872. [ICORE A*]
- [3] W. Wang, M. Xu, Z. Cao, J. Guo, C. Liu, H. Zhang, and X. Zhang. Unified Data Synthesis for Automated 3D Visual Inspection and Digital Twinning of Bridges. *Automation in Construction*, 182:106741, 2026. [JCR Q1]
- [4] M. Xu, P. Ju, J. Liu, and H. Yang. PSMGD: Periodic Stochastic Multi-Gradient Descent for Fast Multi-Objective Optimization. *AAAI 2025*, 39(20):21770–21778. [ICORE A*]
- [5] C. Li, M. Xu, Y. Du, L. Liu, C. Shi, Y. Wang, H. Liu, and Y. Chen. Practical Adversarial Attack on WiFi Sensing Through Unnoticeable Communication Packet Perturbation. *ACM MobiCom 2024*, pp. 373–387. [ICORE A*]

Technical Skills

Programming & tools Python, C/C++, MATLAB, Kotlin; Linux, Git, RT-Thread, Vivado, XSCT/Tcl, $\text{\LaTeX}$

Machine learning PyTorch, TensorFlow; image classification, segmentation, data augmentation

English IELTS Academic: 7.0 overall (Reading: 8.5); Duolingo: 125

Training: Google Data Analytics (2022); Python for Everybody (UMich), Stanford ML, DeepLearning.AI Deep Learning (all 2021).

Industry Experience and Projects

Embedded AI Project: Garbage Classification Device

Jul 2021

NICT Embedded Development and AI Summer School

- Trained a TensorFlow 2 image classifier with data augmentation and deployed it on ART-Pi with RT-AK and RT-Thread; integrated camera/display drivers and bilinear image resizing. [Code]

Design Engineer / Intern (Embedded Software)

Mar 2019 – Jun 2021

Jabil Circuit (Shanghai) Co., Ltd., Wireless R&D

- **Firmware and drivers:** Developed C drivers for RRU EEPROM/power control and integrated them into the HAL; contributed BSP code using I2C, SPI, UART, and DMA.
- **Embedded applications:** Built a C NMEA library and RT-Thread applications for HC-02 Bluetooth and L80 GPS modules.
- **Hardware integration:** Selected MCUs, interpreted schematics, and supported temperature-control testing with hardware engineers using Arduino Uno R3 and the TI LMP92066 DAC.
- **FPGA tools and validation:** Tested Xilinx Zynq PS/PL in Vivado; automated bootloader loading/initialization with XSCT/Tcl and WDB-to-VCD conversion with Python for waveform analysis.

Honors and Awards

2026 ICML Gold Reviewer Award

2025 Fully funded EPSRC postgraduate research scholarship, Swansea University

2022 M.Sc. Certificate of Merit, The Chinese University of Hong Kong

2018 Second Prize, TI Cup Undergraduate Shanghai Electronics Design Contest

2018 First Prize, Service Robot Special Competition, China Robot Competition

2017, 2016 National Encouragement Scholarship, Shanghai University

2016 Self-improvement Scholarship, Shanghai University